

Panel A of Figure 8.4 shows that the short-term correlations of stocks with inflation are low—below 10%. Moving out to four to five years, they peak around 30% and then taper off around 20% at the ten-year horizon. T-bills, however, are a wonderful inflation hedge. At short horizons, below one year, the correlations of T-bills with inflation are above 20% and then increase steadily to around 60% at the ten-year horizon. Because raw stock returns are poor inflation hedges, T-bills are good inflation hedges, and excess stock returns subtract T-bills from stock returns, the excess stock return correlations with inflation are lower than the raw stock correlations with inflation. They are around zero below one year, reach a maximum around 10% around years three to four, and then turn negative around year seven. Clearly, stocks are a poor inflation hedge in terms of the way stock returns co-move with inflation, even though stocks have a high average return.¹⁷

The true picture is even worse than Panel A of Figure 8.4. Panel A uses Pearson, or classical, correlations, which most of the literature and practitioners focus on. Panel B plots Spearman, or robust, rank correlations. Spearman correlations are robust to outliers, which greatly impact the calculations of simple correlations. Panel B shows that some of the correlations in Panel A are spurious and caused by outliers; the robust correlations with inflation are lower than the simple correlation measures in Panel A. Notably the correlations of excess stock returns are now negative for all horizons and after one year, range between -20% and -10%.

The correlation of stock returns and inflation has also changed over time, as shown in Figure 8.5. Figure 8.5 plots the rolling ten-year classical and robust correlations of excess stock returns and inflation. For some of the 1940s and 1950s, the correlation was positive, but from the 1950s it has been negative. Recently, post-2000, the correlation is barely positive (still below 10%), at least measured by simple correlations, but it remains negative after accounting for outliers. While it has changed sign, the correlation of excess stock returns and inflation has never reached above 20% and in fact has gone below -40%.

T-bills are a good inflation hedge, producing three- to five-year (simple and robust) correlations with inflation close to 0.5 because short-term interest rates directly reflect expected inflation. This is through the Fisher hypothesis and through the actions of monetary policy, which we cover in chapter 9. However, a good inflation hedge does not necessarily mean an overall high nominal or real return; a good inflation hedge is an asset that tends to move together with inflation. In terms of returns, T-bills just beat inflation. The average return of T-bills

¹⁷ Ang, Brière, and Signori (2012) also find that within the cross section of equities, it is difficult to find individual stocks that can hedge inflation risk by computing stock inflation betas. Konchitchki (2011) finds more success in finding individual stocks that correlate highly with inflation using accounting data.